Dohyun (Eric) Kim

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EDUCATION

University of Auckland — **Auckland, New Zealand**
Bachelor of Engineering (Honours) in Computer Systems Engineering — *Expected Graduation Nov 2027*

EXPERIENCES

Research Assistant — **Auckland, New Zealand**
University of Auckland — *Jul 2026 - Oct 2026*

- Conducted independent technical study of the **robot soccer system** at UoA's CARES lab, covering camera calibration, **vision processing** and decision strategy design, ahead of migration to a modern framework.
- Investigated **TurtleBot2** navigation algorithms, **LiDAR integration** and control using **ROS** and **Python** to build practical foundations in autonomous systems.

Robotics Instructor — **Auckland, New Zealand**
Creative Imaginary Lab (ciLab) — *Apr 2026 - Oct 2026*

- Instructed **50+ students** in robot **hardware assembly and software programming** to complete missions.
- Supported student development and coached teams in preparation for **nationwide robotics competitions**.

SKILLS

Programming Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL, R, HTML/CSS
Tools & Frameworks: React, Tailwind CSS, Node.js, Express, JUnit, ROS, Android Studio, Figma, REST APIs, Spring Boot
Cloud & DevOps: AWS(Lambda, DynamoDB, S3, Bedrock, Kinesis), Docker, Terraform, Linux, Git, GitHub Actions

PROJECTS

Sentiment PULSE — **Live Demo**
Real-time Sentiment Dashboard - Individual Project — *Jun 2026*

- Built a serverless NLP pipeline with **AWS (Kinesis, Lambda, Comprehend, DynamoDB)** using AWS SAM, with idempotent writes, partial batch response retries, and a dead letter queue **to classify streaming text sentiment in real time** without data loss.
- Designed a **bounded DynamoDB GSI query pattern** instead of full table scans to keep read latency low, and batched Comprehend calls per Lambda invocation to cut API overhead.
- Developed a live React dashboard on AWS Amplify with **CI via GitHub Actions (21 frontend, 8 backend tests)** and a client side simulation fallback, applying cloud and AI/ML fundamentals **from certification study to a hands on project.**

RoastWorks Analytics — **Python**
Business Analytics Dashboard - Team Project — *May 2026*

- Built a PyQt6 desktop analytics app with **pandas and Matplotlib**, automating a manual Excel workflow across **3 business units** and **52,000+ records**, cutting report time from a full day to **under 30 seconds**.
- Implemented three forecasting models across **36 months of data**, combining **scikit-learn** regression with a custom **NumPy** build of Holt's Exponential Smoothing, validated with MAE and RMSE.

AWARD

Top 8 Finalist in AWS×BNZ AI Hackathon 2026 — **Live Demo**
"Spottern!" - Statement Level Fraud Detection Platform — *Jul 2026*

- Built a serverless fraud pipeline with **AWS (SNS, Lambda, Textract, Bedrock, DynamoDB)**, using Claude Opus 4.8 on Bedrock **to extract and flag anomalies from bank statements** with cited transaction evidence.
- **Proposed the approach to BNZ** by drawing on prior individual project experience with serverless AWS pipelines, placing **top 8 out of 20+** teams.

CERTIFICATIONS

HashiCorp Certified: Terraform Associate · HashiCorp — *Sep 2026*
AWS Certified Solutions Architect - Associate · Amazon Web Services — *Aug 2026*
AWS Certified AI Practitioner · Amazon Web Services — *May 2026*
AWS Certified Cloud Practitioner · Amazon Web Services — *Apr 2026*

LEADERSHIP

Academic Team Executive — **Auckland, New Zealand**
KEB (Korean Engineering Body) - Founding Member — *Jul 2024 - Oct 2026*

- **Co-founded** an engineering community of **170+** members; contributed to a **17-member** executive team.
- Delivered **tutorial sessions** for junior engineering students and supported academic event planning.